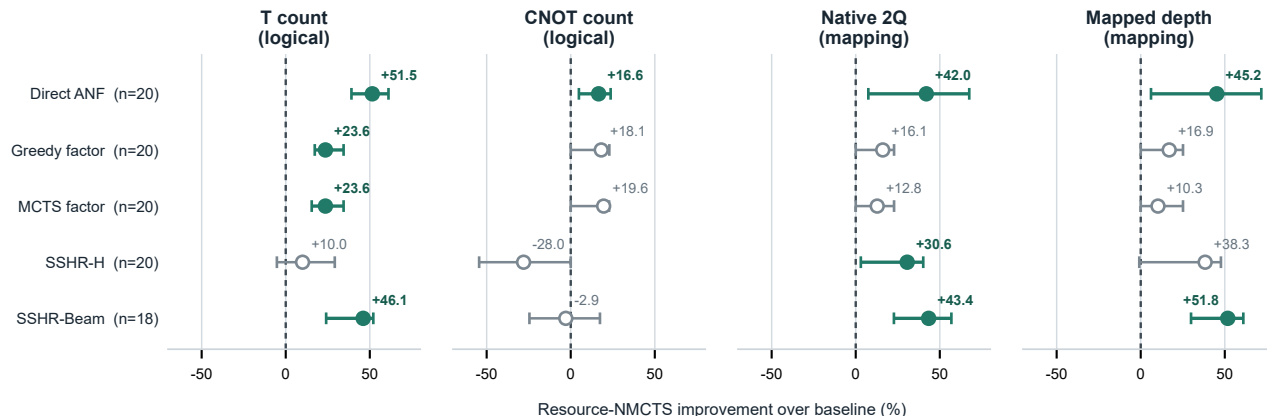


a

Function-level median relative improvement with 95% bootstrap CI

positive favours Resource-NMCTS

● strict support ○ touches/crosses zero or Holm not rejected



b

Inferential audit: W/T/L and family-Holm p

	T count	CNOT count	Native 2Q	Mapped depth
Direct ANF (n=20)	● strict W/T/L 17/3/0 pH 5.82e-4	● strict W/T/L 14/3/3 pH 1.18e-3	● strict W/T/L 14/3/3 pH 1.84e-3	● strict W/T/L 14/3/3 pH 1.84e-3
Greedy factor (n=20)	● strict W/T/L 15/5/0 pH 1.26e-3	○ neutral W/T/L 12/5/3 pH 2.37e-3	○ neutral W/T/L 12/5/3 pH 2.37e-3	○ neutral W/T/L 12/5/3 pH 1.71e-3
MCTS factor (n=20)	● strict W/T/L 15/5/0 pH 1.26e-3	○ neutral W/T/L 12/5/3 pH 2.37e-3	○ neutral W/T/L 12/5/3 pH 2.37e-3	○ neutral W/T/L 12/5/3 pH 1.71e-3
SSHR-H (n=20)	○ neutral W/T/L 10/4/6 pH 0.776	○ neutral W/T/L 5/2/13 pH 0.167	● strict W/T/L 14/2/4 pH 2.58e-3	○ neutral W/T/L 13/1/6 pH 2.58e-3
SSHR-Beam (n=18)	● strict W/T/L 17/1/0 pH 5.78e-4	○ neutral W/T/L 8/1/9 pH 0.831	● strict W/T/L 15/1/2 pH 1.84e-3	● strict W/T/L 15/1/2 pH 1.84e-3

Strict gate: family-Holm reject AND raw median Δ 95% CI upper < 0 AND median relative-improvement 95% CI lower > 0 (zero-touching is neutral).

3 strictly paired seeds per function (7, 17, 29); for SSHR-Beam, all 3 seeds for 2 AES functions hit the 300 s synthesis timeout; hence n=18. pH = family-Holm adjusted p.